WATER QUALITY INDEX (WQI) ASSESSMENT

Damodar River, Dhanbad
Environmental Engineering Project

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1. Objective

To assess the surface water quality of the Damodar River in the Dhanbad coal-mining belt by determining the Water Quality Index (WQI) using the Weighted Arithmetic Index Method, and to compare water quality across three sampling stations representing upstream, near-industrial-discharge, and downstream conditions.

2. Apparatus / Instruments Used

- Digital pH meter
- Dissolved Oxygen (DO) meter / Winkler's titration kit
- BOD incubator (for 5-day BOD test at 20°C)
- TDS meter / conductivity meter
- UV-Visible spectrophotometer (for Nitrate estimation)
- Titration setup (burette, pipette, conical flask) for Chloride and Total Hardness
- Ion-selective electrode / spectrophotometer (for Fluoride)
- Sample bottles (clean, pre-rinsed polyethylene/glass bottles), ice box for BOD sample preservation

3. Procedure

- Three sampling stations were selected along the Damodar River stretch in Dhanbad: **S1 (Upstream)**, **S2 (near an industrial/coal-washery discharge point)**, and **S3 (Downstream)**.
- Grab water samples were collected in pre-cleaned bottles at each station, with separate dark/cold-preserved bottles for BOD samples.
- pH and TDS were measured on-site using calibrated digital meters immediately after collection.
- DO was fixed on-site (Winkler's method) to prevent atmospheric interference before laboratory titration.
- BOD was determined by incubating samples for 5 days at 20°C and measuring the drop in DO (BOD■ method).
- Nitrate, Chloride, Total Hardness, and Fluoride were determined in the laboratory using standard titrimetric/spectrophotometric methods as per APHA Standard Methods.
- Each measured value (Vn) was compared against the corresponding BIS 10500:2012 / CPCB permissible standard (Sn) and ideal value (Vi).
- The Weighted Arithmetic WQI was computed using the unit weight (Wn) and quality rating (Qn) for each parameter, and the water was classified using the standard WQI rating scale.

4. Observations (Measured Data)

Parameter	S1 - Upstream	S2 - Near Discharge	S3 - Downstream	Standard (Sn)	Ideal (Vi)
pH	7.2	7.8	7.5	8.5	7.0
DO (mg/L)	6.8	4.2	5.5	5.0	14.6
BOD (mg/L)	3.2	6.5	4.8	5.0	0
TDS (mg/L)	320	560	440	500	0
Nitrate (mg/L)	15	28	20	45	0
Chloride (mg/L)	90	180	130	250	0
Total Hardness (mg/L)	220	340	280	300	0
Fluoride (mg/L)	0.25	0.42	0.33	1.5	0

Standard (Sn) and Ideal (Vi) values as per BIS 10500:2012 drinking water standards / CPCB guidelines.

5. Calculations (Weighted Arithmetic Method)

Step 1 — Unit Weight:

$$Wn = (1/Sn) / \Sigma(1/Sn)$$

Step 2 — Quality Rating:

$$Qn = |\ (Vn - Vi) / (Sn - Vi)\ | \times 100$$

Step 3 — Water Quality Index:

$$WQI = \Sigma(Qn \times Wn)$$

Sample calculation for Station S2 (Near Industrial Discharge) — BOD parameter:

$Qn(BOD) = |(6.5 - 0)/(5.0 - 0)| \times 100 = 130.0$ | $Wn(BOD) = (1/5.0) / 1.2160 = 0.1645$ | $Qn \times Wn = 21.39$

Parameter	Wn	Qn (S1)	Qn (S2)	Qn (S3)
pH	0.0968	13.33	53.33	33.33
DO	0.1645	81.25	108.33	94.79
BOD	0.1645	64.00	130.00	96.00
TDS	0.0016	64.00	112.00	88.00
Nitrate	0.0183	33.33	62.22	44.44
Chloride	0.0033	36.00	72.00	52.00
Hardness	0.0027	73.33	113.33	93.33

Fluoride	0.5483	16.67	28.00	22.00

Station	WQI	Classification
S1 - Upstream	**35.35**	**Good**
S2 - Near Industrial Discharge	**61.59**	**Poor**
S3 - Downstream	**48.06**	**Good**

6. Results and Graphical Analysis

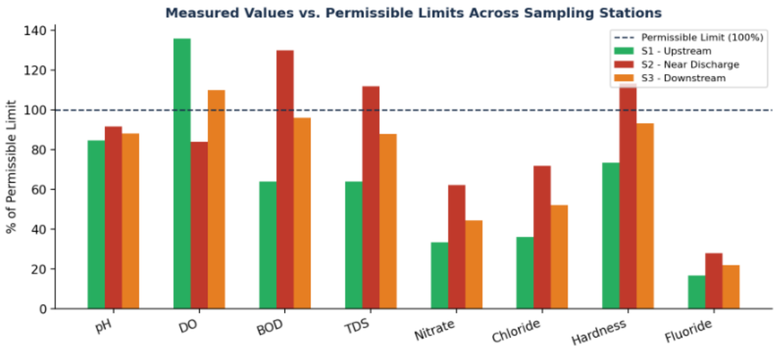

Fig. 1 — Measured parameter values (as % of permissible limit) across the three stations. Bars crossing the 100% line exceed the CPCB/BIS permissible standard.

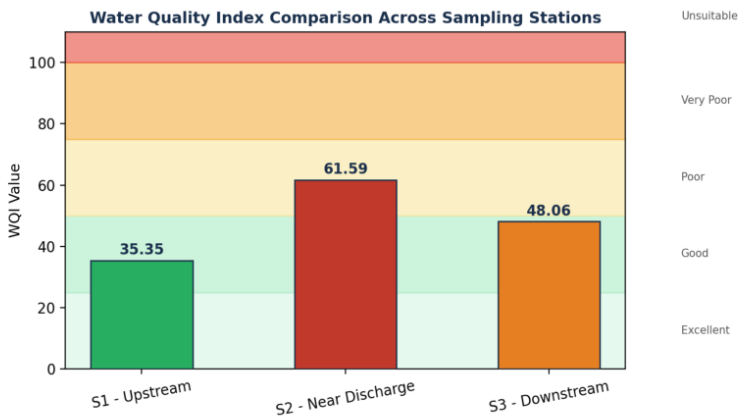

Fig. 2 — WQI comparison across sampling stations, overlaid on the standard classification bands.

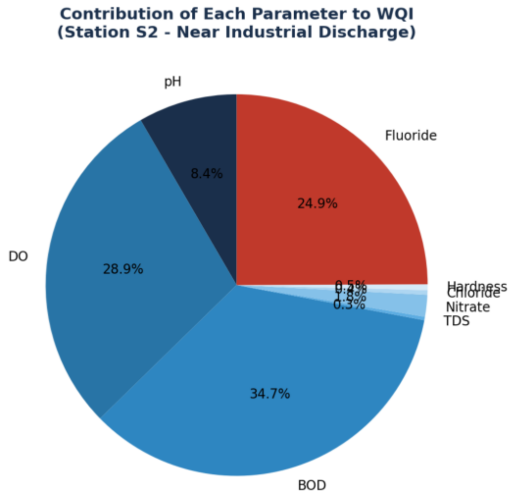

Fig. 3 — Percentage contribution of each parameter to the WQI at Station S2 (worst-affected site). Fluoride and BOD/DO dominate the index due to their low permissible limits and elevated concentrations.

7. Conclusion and Recommendations

The WQI values obtained follow a clear pollution gradient: water quality is **"Good"** upstream (WQI = 35.35), deteriorates to **"Poor"** near the industrial/coal-washery discharge point (WQI = 61.59), and partially recovers downstream to **"Good"** (WQI = 48.06) due to natural dilution and self-purification of the river.

BOD, DO, and Fluoride were found to be the dominant contributors to the elevated WQI at the discharge point, indicating organic pollution and oxygen depletion consistent with untreated industrial/domestic effluent discharge — a pattern widely reported in published studies of the Damodar River in the Dhanbad coal-mining belt.

Recommendations: (i) Treatment of industrial effluent (coal washery/mining discharge) before release into the river; (ii) regular monitoring of BOD and DO near discharge points; (iii) enforcement of CPCB effluent discharge norms; and (iv) seasonal WQI monitoring (pre-monsoon, monsoon, post-monsoon) to capture temporal variation in pollution load.